



**PETRON**  
**MALAYSIA**

Product Name: LIQUEFIED PETROLEUM GAS  
Revision Date: 20 May 2015  
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## SAFETY DATA SHEET

### SECTION 1 CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

As of the revision date above, this SDS meets the regulations in Malaysia.

#### PRODUCT

**Product Name:** LIQUEFIED PETROLEUM GAS  
**Product Description:** Liquefied Hydrocarbon Gas, Gas or Liquefied Gas  
**Product Code:** 249813-88  
**Intended Use:** Fuel for household and industrial applications

| Trade Names | Trade Names  |
|-------------|--------------|
| LPG         | PETRON GASUL |

#### COMPANY IDENTIFICATION

**Supplier:** Petron Subsidiaries in Malaysia  
Menara I&P, Tower 1,  
No. 46, Jalan Dungun, Damansara Heights,  
50490 Kuala Lumpur, Malaysia

**Supplier General Contact** +603 2082 8400

### SECTION 2 HAZARDS IDENTIFICATION

**GHS CLASSIFICATION** Gases Under Pressure, Liquefied Gas  
Flammable Gases – Category 1

#### GHS Labeling

#### Symbol(s)



**Signal Word** Danger

#### Hazard Statements

H220 Extremely flammable gas  
H280 Contains gas under pressure; may explode if heated  
H316 Causes mild skin irritation  
H320 Causes eye irritation  
H336 May cause drowsiness and dizziness



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## PRECAUTIONARY STATEMENTS

### General Precautionary Statements

P101 If medical advice is needed, have product container or label at hand  
P102 Keep out of reach of children  
P103 Read label before use

### Prevention Precautionary Statements

P264 Wash hands thoroughly after handling  
P261 Avoid breathing fume/gas/mist/vapor  
P271 Use only in a well-ventilated area  
P273 Avoid release to the environment

### Response Precautionary Statements

P312 Call a Poison Center or Physician if you feel unwell  
P304 + P312 IF INHALED: Call a Poison Center or Physician if you feel unwell  
P332 + P313 If skin irritation occurs: get medical advice/attention  
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove lenses, if present and easy to do. Continue rinsing  
  
P337 + P313 If eye irritation persists: Get medical advice / attention  
P304 + P340 + P312 IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing. Call a Poison Center or Physician if you feel unwell

### Storage Precautionary Statements

P410 + P403 Protect from sunlight, Store in a well-ventilated place  
P405 Store locked up

### Disposal Precautionary Statement

P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

## SECTION 3

## COMPOSITION / INFORMATION ON INGREDIENTS

This material is regulated as a complex substance.

### Reportable Hazardous Substance(s) or Complex Substance(s)

| Name                    | CAS#       | Concentration* |
|-------------------------|------------|----------------|
| LIQUEFIED PETROLEUM GAS | 68476-85-7 | 100 %          |

### Reportable Hazardous Constituent(s) Contained in Complex Substance(s)

| Name          | CAS#     | Concentration* |
|---------------|----------|----------------|
| 1,3-BUTADIENE | 106-99-0 | < 0.5%         |



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\* All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.

#### **SECTION 4 FIRST AID MEASURES**

##### **INHALATION**

Immediately remove from further exposure. Get immediate medical assistance. For those providing assistance, avoid exposure to yourself or others. Use adequate respiratory protection. Give supplemental oxygen, if available. If breathing has stopped, assist ventilation with a mechanical device or use mouth-to-mouth resuscitation.

##### **SKIN CONTACT**

If product is injected into or under the skin, or into any part of the body, regardless of the appearance of the wound or its size, the individual should be evaluated immediately by a physician as a surgical emergency. Even though initial symptoms from high pressure injection may be minimal or absent, early surgical treatment within the first few hours may significantly reduce the ultimate extent of injury. If frostbite occurs, immerse involved area in water at body temperature. Keep immersed for 20 to 40 minutes. Seek medical assistance.

##### **EYE CONTACT**

Flush thoroughly with water for at least 15 minutes. Get medical assistance.

##### **INGESTION**

Not Applicable

#### **SECTION 5 FIRE FIGHTING MEASURES**

##### **EXTINGUISHING MEDIA**

###### **Appropriate Extinguishing Media:**

Use water fog, dry chemical or carbon dioxide (CO<sub>2</sub>) to extinguish flames.

###### **Inappropriate Extinguishing Media:**

Straight streams of water

##### **FIRE FIGHTING**

###### **Fire Fighting Instructions:**

Allow the fire to burn under controlled conditions. Stop leak if you can do so without risk. Evacuate area. If a leak or spill has not ignited, use water spray to disperse the vapours and to protect personnel attempting to stop a leak. Prevent run-off from fire control or dilution from entering streams, sewers or drinking water supply. Fire-fighters should use standard protective equipment and in enclosed spaces, self-contained breathing apparatus (SCBA). Use water spray to cool fire exposed surfaces and to protect personnel.

###### **Unusual Fire Hazards:**

Flammable Gas. Hazardous material. Firefighters should



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consider protective equipment indicated in Section 8.

**Hazardous Combustion Products:**

Incomplete combustion products, Oxides of carbon

**FLAMMABILITY PROPERTIES**

**Flash Point, Closed Cup:**

<0°C (32°F) [Estimated]

**Flammable Limits**

**(Approximate volume % in air):**

LEL: 2.0 UEL: 9.5

**Auto-ignition Temperature:**

>250°C (482°F)

**SECTION 6**

**ACCIDENTAL RELEASE MEASURES**

**NOTIFICATION PROCEDURES**

In the event of a spill or accidental release, notify relevant authorities in accordance with all applicable regulations.

**PROTECTIVE MEASURES**

Avoid contact with spilled material. Warn or evacuate occupants in surrounding and downwind areas if required, due to toxicity or flammability of the material. See Section 5 for firefighting information. See the Hazard Identification Section for Significant Hazards. See Section 4 for First Aid Advice. See Section 8 for advice on the minimum requirements for personal protective equipment. Additional protective measures may be necessary, depending on the specific circumstances and/or the expert judgment of the emergency responders.

**SPILL MANAGEMENT**

**Land Spill:**

Eliminate all ignition sources (no smoking, flares, sparks or flames in immediate area). Stop leak if you can do so without risk. CAUTION: When in contact with refrigerated/cryogenic liquids, many materials become brittle and are likely to break without warning. Allow liquid to evaporate from the surface. Use water spray to reduce vapour or divert vapour cloud drift. All equipment used when handling the product must be grounded. Do not direct water at spill or source of leak. Do not touch or walk through spilled material. If possible, turn leaking containers so that gas escapes rather than liquid. Isolate area until gas has dispersed. Prevent spreading of vapour through sewers, ventilation systems and confined areas.

**Water Spill:**

Eliminate all ignition sources (no smoking, flares, sparks or flames in immediate area). Do not confine in area of spill. Allow liquid to evaporate from the surface. See Land Spill section of the SDS for advice on gases.



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Water spill and land spill recommendations are based on the most likely spill scenario for this material; However, geographic conditions, wind, temperature, (and in the case of a water spill) wave, and current direction, and speed may greatly influence the appropriate action to be taken. For this reason, local experts should be consulted,

Note: Local regulations may prescribe or limit action to be taken.

#### **ENVIRONMENTAL PRECAUTIONS**

Prevent entry into waterways, sewers, basements or confined areas.

### **SECTION 6**

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## ENVIRONMENTAL PRECAUTIONS

Prevent entry into waterways, sewers, basements or confined areas.

## SECTION 7 HANDLING AND STORAGE

### HANDLING

Avoid all personal contact. Prevent exposure to ignition sources, for example use non-sparking tools and explosion-proof equipment. Do not enter storage areas or confined spaces unless adequately ventilated. Use proper bonding and/or earthing procedures. However, bonding and earthing may not eliminate the hazard from static accumulation. Material can accumulate static charges which may cause an electrical spark (ignition source).

### Static Accumulator:

This material is a static accumulator.

### STORAGE

Ample fire water supply should be available. A fixed sprinkler/deluge system is recommended. The container choice, for example storage vessel, may effect static accumulation and dissipation. Keep container closed. Handle containers with care. Open slowly in order to control possible pressure release. Store in a cool, well-ventilated area. Outside or detached storage preferred. Storage containers should be earthed and bonded.

## SECTION 8 EXPOSURE CONTROLS / PERSONAL PROTECTION

### Exposure limits/standards (Note: Exposure limits are not additive)

| Substance Name          | Form | Limit/Standard |                        |          | Note | Source       | Year |
|-------------------------|------|----------------|------------------------|----------|------|--------------|------|
| 1,3-BUTADIENE           |      | PEL            | 4.4 mg/m <sup>3</sup>  | 2 ppm    |      | Malaysia PEL | 2000 |
| 1,3-BUTADIENE           |      | TWA            | 2 ppm                  |          |      | ACGIH        | 2011 |
| LIQUEFIED PETROLEUM GAS |      | PEL            | 1800 mg/m <sup>3</sup> | 1000 ppm |      | Malaysia PEL | 2000 |
| LIQUEFIED PETROLEUM GAS |      | TWA            | 1000 ppm               |          |      | ACGIH        | 2011 |

NOTE: Limits/standards shown for guidance only. Follow applicable regulations.

### ENGINEERING CONTROLS

The level of protection and types of controls necessary will vary depending upon potential exposure conditions. Control measures to consider.

Use explosion-proof ventilation equipment to stay below exposure limits.



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## **PERSONAL PROTECTION**

Personal protective equipment selections vary based on potential exposure conditions such as applications, handling practices, concentration and ventilation. Information on the selection of protective equipment for use with this material, as provided below, is based upon intended, normal usage.

### **Respiratory Protection:**

If engineering controls do not maintain airborne contaminant concentrations at a level which is adequate to protect worker health, an approved respirator may be appropriate. Respirator selection, use, and maintenance must be in accordance with regulatory requirements, if applicable. Types of respirators to be considered for this material include: No special requirements under ordinary conditions of use and with adequate ventilation.

For high airborne concentrations, use an approved supplied-air respirator, operated in positive pressure mode. Supplied air respirators with an escape bottle may be appropriate when oxygen levels are inadequate, gas/vapour warning properties are poor, or if air purifying filter capacity/rating may be exceeded.

### **Hand Protection:**

Any specific glove information provided is based on published literature and glove manufacturer data. Glove suitability and breakthrough time will differ depending on the specific use conditions. Contact the glove manufacturer for specific advice on glove selection and breakthrough times for your use conditions. Inspect and replace worn or damaged gloves. The types of gloves to be considered for this material include: Thermally protective, chemical resistant gloves are recommended. If contact with forearms is likely, wear gauntlet-style gloves. Nitrile, Viton

### **Eye Protection:**

Face shield is recommended.

### **Skin and Body Protection:**

Any specific clothing information provided is based on published literature or manufacturer data. The types of clothing to be considered for this material include:

Thermally protective and chemical resistant apron and long sleeves are recommended when volume of material is significant.

### **Specific Hygiene Measures:**

Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Discard contaminated clothing and footwear that cannot be cleaned. Practice good housekeeping.



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**Note:** Physical and chemical properties are provided for safety, health and environmental considerations only and may not fully represent product specifications. Contact the Supplier for additional information.

#### GENERAL INFORMATION

|                  |                            |
|------------------|----------------------------|
| Physical State:  | Gas                        |
| Form:            | Liquified                  |
| Colour:          | Colourless                 |
| Odour:           | Odorless (may be odorized) |
| Odour Threshold: | N/A                        |

#### IMPORTANT HEALTH, SAFETY, AND ENVIRONMENTAL INFORMATION

|                                          |                                            |
|------------------------------------------|--------------------------------------------|
| Relative Density:                        | 0.5300 - 0.5800                            |
| Flash Point :                            | <0°C (32°F) [Estimated]                    |
| Flammable Limits                         |                                            |
| (Approximate volume % in air):           | LEL: 2.0 UEL: 9.5                          |
| Auto-ignition Temperature:               | >250°C (482°F)                             |
| Boiling Point / Range:                   | < 0°C (32°F)                               |
| Vapour Density (Air = 1):                | 0.70 @ 101 kPa - 1.1 at 101 kPa            |
| Vapour Pressure:                         | > 133 kPa (1000 mm Hg) at 20°C [Estimated] |
| Evaporation Rate (n-butyl acetate = 1):  | N/D                                        |
| pH:                                      | N/A                                        |
| Log Pow                                  |                                            |
| (n-Octanol/Water Partition Coefficient): | N/A                                        |
| Solubility in Water:                     | Negligible                                 |
| Viscosity:                               | N/A                                        |
| Oxidizing Properties:                    | See Hazards Identification Section.        |

#### OTHER INFORMATION

|                 |     |
|-----------------|-----|
| Freezing Point: | N/D |
| Melting Point:  | N/D |

| SECTION 10 | STABILITY AND REACTIVITY |
|------------|--------------------------|
|------------|--------------------------|

|                                   |                                                             |
|-----------------------------------|-------------------------------------------------------------|
| STABILITY:                        | Material is stable under normal conditions.                 |
| CONDITIONS TO AVOID:              | Avoid heat, sparks, open flames and other ignition sources. |
| MATERIALS TO AVOID:               | Strong oxidizers                                            |
| HAZARDOUS DECOMPOSITION PRODUCTS: | Material does not decompose at ambient temperatures.        |
| HAZARDOUS POLYMERIZATION:         | Will not occur.                                             |

| SECTION 11 | TOXICOLOGICAL INFORMATION |
|------------|---------------------------|
|------------|---------------------------|

#### ACUTE TOXICITY



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| <u>Route of Exposure</u>          | <u>Conclusion / Remarks</u>                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                 |                                                                                                   |
| Toxicity (Rat): LC50 > 5000 mg/m3 | Minimally Toxic. Based on assessment of the components.                                           |
| Irritation: No end point data.    | Negligible hazard at ambient/normal handling temperatures. Based on assessment of the components. |
|                                   |                                                                                                   |
| <b>Ingestion</b>                  |                                                                                                   |
| Toxicity: No end point data.      | Not applicable.                                                                                   |
|                                   |                                                                                                   |
| <b>Skin</b>                       |                                                                                                   |
| Toxicity: No end point data.      | Not applicable.                                                                                   |
| Irritation: No end point data.    | Negligible irritation to skin at ambient temperatures. Based on assessment of the components.     |
|                                   |                                                                                                   |
| <b>Eye</b>                        |                                                                                                   |
| Irritation: No end point data.    | May cause mild, short-lasting discomfort to eyes. Based on assessment of the components.          |

## CHRONIC/OTHER EFFECTS

### For the product itself

May cause central nervous system disorder (e.g., narcosis involving a loss of coordination, weakness, fatigue, mental confusion and blurred vision) and/or damage. Exposure to rapidly expanding gas or vaporizing liquid may cause frostbite (cold burn). Simple asphyxiant: Acts by displacing oxygen in the lungs thereby diminishing the supply of oxygen available to the blood and tissues. Symptoms include shortness of breath, rapid heart rate, incoordination, lethargy, headaches, nausea, vomiting, and disorientation. Continued lack of oxygen may result in convulsions, loss of consciousness and death. Since exercise increases the tissue need for oxygen, symptoms will occur more quickly during exertion in an oxygen-deficient environment. Oxygen in enclosed spaces should be maintained at 21 percent by volume.

### Contains:

1,3- Butadiene. 1,3-Butadiene is a multi-site carcinogen in rodents. Epidemiology studies indicate an association between exposure to 1,3-butadiene and leukemia in humans. Mutations have been observed in in-vitro and in-vivo rodent assays. Although several older studies had conflicting results, a newer screening study in rats showed no adverse reproductive or developmental effects.

Additional information is available by request.



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#### **IARC Classification:**

The following ingredients are cited on the lists below:

| Chemical Name | CAS Number | List Citations |
|---------------|------------|----------------|
| 1,3-BUTADIENE | 106-99-0   | 1              |

--REGULATORY LISTS SEARCHED--

1 = IARC 1

2 = IARC 2A

3 = IARC 2B

### **SECTION 12 ECOLOGICAL INFORMATION**

The information given is based on data available for the material, the components of the material, and similar materials.

**ECOTOXICITY** Material -- Not expected to be harmful to aquatic organisms.

**MOBILITY** Material -- Highly volatile, will partition rapidly to air. Not expected to partition to sediment and wastewater solids.

#### **PERSISTENCE AND DEGRADABILITY**

**Biodegradation:** Material -- Expected to be inherently biodegradable

**Atmospheric Oxidation:** Material -- Expected to degrade at a moderate rate in air

**BIOACCUMULATION POTENTIAL** Material -- Potential to bioaccumulate is low.

### **SECTION 13 DISPOSAL INFORMATION**

Disposal recommendations based on material as supplied. Disposal must be in accordance with current applicable laws and regulations, and material characteristics at time of disposal.

**DISPOSAL RECOMMENDATIONS** Product is suitable for burning in an enclosed controlled burner for fuel value or disposal by supervised incineration at very high temperatures to prevent formation of undesirable combustion products.

**Empty Container Warning** Empty Container Warning (where applicable): Empty containers may contain residue and can be dangerous. Do not attempt to refill or clean containers without proper instructions. Empty drums should be completely drained and safely stored until appropriately reconditioned or disposed. Empty containers should be taken for recycling, recovery, or disposal through suitably qualified or licensed contractor and in accordance with governmental regulations. Do not Pressurize, cut, weld, braze,



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solder, drill, grind, or expose such containers to heat, flame, sparks, static electricity, or other sources of ignition. They may explode and cause injury or death.

#### SECTION 14

#### TRANSPORT INFORMATION

##### LAND

**Proper Shipping Name:** PETROLEUM GASES, LIQUEFIED  
**Hazard Class:** 2  
**Hazchem Code:** 2YE  
**UN Number:** 1075  
**Packing Group:** (N/A)  
**Label(s) / Mark(s):** 2.1

##### SEA (IMDG)

**Proper Shipping Name:** PETROLEUM GASES, LIQUEFIED  
**Hazard Class & Division:** 2.1  
**EMS Number:** F-D, S-U  
**UN Number:** 1075  
**Packing Group:** (N/A)  
**Label(s):** 2.1  
**Transport Document Name:** UN1075, PETROLEUM GASES, LIQUEFIED, 2.1, (0°C c.c.)

##### AIR (IATA)

**Proper Shipping Name:** PETROLEUM GASES, LIQUEFIED  
**Hazard Class & Division:** 2.1  
**UN Number:** 1075  
  
**Packing Group:** (N/A)  
**Label(s) / Mark(s):** 2.1  
**Transportation Limitations:** CARGO AIRCRAFT ONLY  
**Transport Document Name:** UN1075, PETROLEUM GASES, LIQUEFIED, 2.1

##### Label

**Flammable**



#### SECTION 15

#### REGULATORY INFORMATION

Material is hazardous as defined by the Occupational Safety and Health (Classification, Packaging, and Labeling of Hazardous Chemicals) Regulations.

#### REGULATORY STATUS AND APPLICABLE LAWS AND REGULATIONS

Complies with the following national/regional chemical inventory requirements: AICS, TSCA,



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IECSC, KECI, PICCS, ENCS, EINECS, DSL

|                   |                          |
|-------------------|--------------------------|
| <b>SECTION 16</b> | <b>OTHER INFORMATION</b> |
|-------------------|--------------------------|

**N/D = Not determined, N/A = Not applicable**

**THIS SAFETY DATA SHEET CONTAINS THE FOLLOWING REVISIONS: 2**

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The information and recommendations contained herein are based on the data sheet and sources believed to be accurate and correct and Petron makes no expressed or implied warranty with respect to the accuracy of the information or the suitability of the recommendations contained in the MSDS. It is your responsibility to contact Petron to ensure that this document is the most current available from Petron. The information and recommendations are offered for the user's consideration and examination. It is the user's full responsibility to satisfy itself that the product is suitable for the intended use and to ensure the safe handling and use of the product. If the user repackages this product, it is the user's responsibility to ensure proper health, safety and other necessary information is included with and/or on the container. Appropriate warnings and safe-handling procedures should be provided to handlers and users. Alteration of this document is strictly prohibited. Petron accepts no responsibility and hereby disclaim all liability for any harmful effects which may be caused by purchase, resale, use, exposure or disposal of the product. Users must comply with all local laws and regulations. Except to the extent required by law, re-publication or retransmission of this document, in whole or in part, is not permitted. The term, "Petron" is used for convenience, and may include any one or more of Petron Corporation, Petron Oil & Gas International Sdn Bhd or any affiliates in which they directly or indirectly hold any interest. In no event, shall Petron or its suppliers be liable for any direct or indirect loss of profits or for any other special, indirect or consequential damages, nor shall the liability of Petron on any claim for damages arising out of or connected with the sale, delivery or use of the product purchased exceed the purchase price of the product.